

A seven-term bound and local lifting obstructions for degree-twenty Casas–Alvero polynomials

Scientific manuscript for review

24 September 2026

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1 Abstract

We prove that a nontrivial characteristic-zero Casas–Alvero polynomial of degree twenty has at least seven nonzero terms after centering at the common root of its nineteenth derivative. The proof reduces small supports to five families and excludes the last family through a complete characteristic-thirteen residue classification and explicit lift obstructions. The specialization and lifting arguments include coefficient de-generation and ramified extensions. We exclude six further exact seven-term supports, reducing the audited seven-term frontier from fourteen possibilities to eight. We also give a multiplicity-preserving factorization of a local resultant norm into marked obstruction values. The latter identifies, but does not resolve, the remaining nonvanishing problem. Exact finite certificates, independent arithmetic checks, and reproducible code accompany the proofs. Neither the unrestricted degree-twenty case nor the general Casas–Alvero conjecture is proved.

2 Introduction

A polynomial over a field of characteristic zero is a Casas–Alvero polynomial if it has a common zero with each of its derivatives of orders one through one less than its degree. The Casas–Alvero conjecture asserts that every such polynomial is a power of a linear polynomial. This paper concerns coefficient supports in degree twenty. It does not prove the conjecture in that degree or in arbitrary degree.

Translate the zero of the nineteenth derivative to zero and make the polynomial monic. The resulting centered polynomial has the form

$$f(X) = X^{20} + \sum_{j \in S} c_j X^{20-j}, \quad S \subseteq \{2, \dots, 19\}, \quad c_j \neq 0.$$

The absence of the constant term follows from the Casas–Alvero condition for the nineteenth derivative. We count the leading monomial, so the total number of terms is $|S| + 1$. A bound on this number concerns the centered polynomial, not all of its translates.

Our principal result is that a nontrivial centered degree-twenty Casas–Alvero polynomial has at least seven terms. A support sieve leaves five exact six-term families. Four admit relatively short exclusions. The remaining family

$$X^{20} + AX^{16} + BX^{15} + CX^{10} + DX^3 + EX, \quad ABCDE \neq 0,$$

is excluded by a complete characteristic-thirteen residue classification and explicit lift obstructions. The argument includes residue-zero nonzero coefficients and ramified ambient extensions; neither may be discarded in a reduction argument. The detailed proof below supplies the finite identities and the route from them to the characteristic-zero conclusion.

Earlier work supplies much of the framework. The prime-adic constraints and determinant restrictions of Castryck, Laterveer and Ounaies [CLO], the sparse criteria of de Frutos Marín [deFrutos], and the shared-derivative restrictions in Massri [Massri] provide the predecessor context. The proof below identifies which restrictions it actually imports; Massri’s additional support filter is redundant in the final small-support sieve. Their combination already implies the weaker five-term bound; we do not claim that bound as new. The singleton and two-visible-coefficient criteria are likewise older results. Marashdeh [Marashdeh] gives related support and triangular-elimination methods. Our new burden is the specific exclusions beyond those criteria, not a claim to have introduced valuation methods or Hensel lifting.

The final six-term family is not eliminated directly by the older singleton/two-visible-coefficient test at any prime: at primes 2, 3, 5, 17, 19, an allowed singleton degeneration has binomial coefficient congruent to one; at primes 7, 11, 13, at least four positions are visible; and at every prime greater than twenty, all five are visible. This explains why inspecting only the finite-field support is insufficient. The lift calculation uses derivative information lost by reduction.

We also record a further reduction of the seven-term support frontier, and a marked factorization theorem for the local resultant algebra of one characteristic-seventeen branch. The latter preserves multiplicities and identifies the remaining scalar obstruction, but does not prove that it is always nonzero. The previously certified exclusions of 79 of 240 canonical supports in that branch remain partial coverage. No surviving finite-field configuration is asserted to lift to a Casas–Alvero polynomial.

The evidence archive distinguishes the final proof path, executable finite checks, historical exploratory material, and external review records. The paper is a computer-assisted mathematical manuscript for review, not a proof-assistant-certified result. Literature comparison found no direct predecessor of the seven-term theorem in the inspected sources, but two specific overlap questions remain unresolved: the exact systems behind ProofAtlas’s reported degree-twenty work and the unavailable full text of Shih, Cheng-Pang’s 2022 thesis. These limits are documented rather than treated as novelty clearance.

3 A seven-term bound for centered degree-20 Casas–Alvero polynomials

This section gives one proof path for the existing sparsity theorem. The finite identities needed in that proof are specified below, including their defining polynomials, coefficient-zero charts, verification bounds, and supplemental files. Earlier dossiers are not additional mathematical assumptions. The result concerns a restricted class of hypothetical counterexamples; it does not settle degree 20 or the Casas–Alvero conjecture.

3.1 S.1. Statement, conventions, and specialization

For a polynomial $f = \sum c_i X^i$, write

$$H_k f = \sum_{i \geq k} \binom{i}{k} c_i X^{i-k}.$$

In characteristic zero, $f^{(k)} = k! H_k f$, so the ordinary and Hasse common-root conditions agree. Say that f has the CA property if f and $H_k f$ have a common root for every $1 \leq k < \deg f$. Roots are taken in an algebraic closure. The polynomial is nontrivial if it is not a scalar multiple of a power of a linear polynomial. Let α be the unique root of $H_{19} f$. The CA property implies $f(\alpha) = 0$. Translate α to zero and make the polynomial monic. Its centered form is

$$F(X) = X^{20} + \sum_{j \in S} c_j X^{20-j}, \quad S \subseteq \{2, \dots, 19\}, \quad c_j \neq 0. \quad (\text{S1})$$

The indices j are **deficiencies**, not exponents or derivative orders. There are exactly $1 + |S|$ nonzero monomials. Changing scale $X \mapsto rX$, $r \neq 0$, does not change S .

Theorem S. Every nontrivial characteristic-zero degree-20 polynomial with the CA property has at least seven nonzero monomials in its centered form.

We first record the specialization argument used throughout. Extend the p -adic valuation to a field containing the coefficients and all roots. Choose a nonzero root ρ of minimum valuation, and replace $F(X)$ by $\rho^{-20} F(\rho X)$. All roots are integral and a root equals 1. In binomial normalization write

$$F(X) = \sum_{j=0}^{20} \binom{20}{j} a_j X^{20-j}, \quad a_0 = 1, \quad a_1 = a_{20} = 0.$$

The monic normalized derivative of degree j is

$$G_j(X) = \frac{H_{20-j}F(X)}{\binom{20}{j}} = \sum_{i=0}^j \binom{j}{i} a_i X^{j-i}.$$

At an integral common root β_j , its equation expresses a_j as an integral polynomial in $a_0, \dots, a_{j-1}, \beta_j$. Induction proves a_j integral. Therefore reduction deletes every ordinary coefficient whose binomial multiplier is divisible by p . It preserves common-root witnesses for the Hasse derivatives and retains a nonzero root. In particular, the reduction is not X^{20} .

No assertion about an arbitrary affine special fiber is used here: integrality and the retained unit root are established before reduction. The argument permits ramification, coefficient degeneration, and coincident witnesses. A valuation extension of the coefficient field is sufficient; the elementary minimum-valuation arguments below also work in an ordered value group. Alternatively, for this finite algebraic existence question one may first specialize an assumed exact-support solution to an algebraic characteristic-zero solution, encoding the nonzero coefficients by inverse variables, and then work over a finite extension of $\mathbf{Q}_p$.

3.2 S.2. Published restrictions and the complete finite support step

The proof uses two published arithmetic restrictions. The first is reproduced here as an elementary support test. The second is the determinant theorem of Castryck, Laterveer, and Ounaies.

For a prime p , put

$$V_p = \{j \in \{2, \dots, 19\} : p \nmid \binom{20}{j}\}.$$

The normalization above shows that $S \cap V_p \neq \emptyset$. If $S \cap V_p = \{r\}$, the reduction is $X^{20} + aX^{20-r}$, with $a \neq 0$. A common root with H_{20-r} is nonzero, and the root and derivative equations imply

$$\binom{20}{r} \equiv 1 \pmod{p}. \quad (\text{S2})$$

This test includes coefficient loss: the remaining visible coefficient cannot also disappear because a unit root was retained.

For completeness, the two-visible test is stated with the necessary degeneration guards. It is the deficiency-index form of de Frutos Marín's two-support criterion [deFrutos]: *Perspectivas aritméticas para la Conjetura de Casas-Alvero*, Theorem 3.5.1 and Proposition 3.5.5, printed pp. 55–57 ([thesis and repository record](#), DOI 10.35376/10324/3602). If $S \cap V_p = \{r, s\}$, $r < s$, define in $\mathbf{F}_p$

$$B = \binom{20}{r}, \quad D = \binom{20}{s}, \quad C = \binom{20-r}{s-r}, \quad g = \gcd(r, s),$$

$$N = B^{s/g}(C-1)^{(s-r)/g}(D-C)^{r/g} - (B-1)^{r/g}(D-1)^{s/g}. \quad (\text{S3})$$

If $B, D \notin \{0, 1\}$, a necessary condition is $N = 0$. The guard excludes singleton degenerations; it must not be omitted.

Here is a direct derivation. Both reduced coefficients must be nonzero by (S2). Normalize the common root for order $20 - r$ to 1. The two coefficients then become $-B$ and $B - 1$. For a nonzero common root v for order $20 - s$, subtraction of the root and derivative equations gives

$$(D - C)v^s = (C - 1)(B - 1), \quad B(D - C)v^{s-r} = (D - 1)(B - 1).$$

The second equality rules out $D = C$, and the first then rules out $C = 1$. Taking powers after division gives (S3). Thus this finite support test can be checked without relying on an unquoted source formula.

The other imported restriction is [Castryck–Laterveer–Ounaies, Theorem 2](#) [CLO]. For degree $p + 1 = 20$, the centered root is simple, and, with

$$Z = \{j : 2 \leq j \leq 18, j \notin S\},$$

the following determinant is zero modulo 19:

$$\Delta(Z) = \det \begin{pmatrix} [-1, (j \binom{j-2}{k-2} \mathbf{1}_{k \leq j})_{k \in Z}]_{j \in Z} \\ -1, ((-1)^k)_{k \in Z} \end{pmatrix}. \quad (\text{S4})$$

The rows and columns indexed by Z are in increasing order. The indices in this formula are the **missing deficiency indices**. In particular, simplicity gives $19 \in S$. This is the only determinant theorem imported here.

The elementary tests already give four disjoint required sets:

$$\{19\}, \quad \{4, 16\}, \quad \{5, 10, 15\}, \quad \{2, 3, 17, 18\}. \quad (\text{S5})$$

They come from primes 19, 2, 5, 17, respectively; at 17 a singleton at 19 is forbidden by $20 \not\equiv 1$. Hence $|S| \geq 4$, without importing an earlier sparsity result.

Apply (S2) and (S3) at $p = 2, 3, 5, 7, 11, 13, 17, 19$, then (S4), to all subsets of $\{2, \dots, 19\}$ of size 4 or 5. This is the entire finite support calculation:

Number of nonleading terms	All subsets	After visible/singleton tests	After guarded two-visible test	After determinant
4	3060	8	4	1
5	8568	100	54	5

The size-four survivor is $\{4, 10, 17, 19\}$. The five size-five survivors, named here once and for all, are:

Family	Deficiency support S	Ordinary exponent support
A	$\{3, 4, 10, 18, 19\}$	$\{20, 17, 16, 10, 2, 1\}$
B	$\{3, 10, 16, 17, 19\}$	$\{20, 17, 10, 4, 3, 1\}$
C	$\{4, 5, 10, 17, 19\}$	$\{20, 16, 15, 10, 3, 1\}$
D	$\{4, 10, 12, 17, 19\}$	$\{20, 16, 10, 8, 3, 1\}$
E	$\{8, 10, 16, 17, 19\}$	$\{20, 12, 10, 4, 3, 1\}$

All operations in this sieve are explicit small integer or finite-field operations in (S2)–(S4). Supplemental code verifies the determinants both by integer Bareiss elimination and modular Gaussian elimination; the degree-12 example printed after CLO Theorem 2 is an indexing control. No search for polynomial coefficients enters this step.

The archived implementation additionally applies the restrictions $S \cap \{5, 10\} \neq \emptyset$ and $S \cap \{10, 15\} \neq \emptyset$, available from the degree-20 characteristic-5 calculation in the proof of [Massri, Theorem 7.9](#) [Massri]. They remove no support after the preceding two-visible tests: the counts remain 4 and 54. They can therefore be omitted from this proof dependency. Neither Massri’s three-recycled-root theorem nor any claimed full proof of the conjecture is needed.

3.3 S.3. The characteristic-13 lemma excluding D and the size-four survivor

Lemma S.3. Over any algebraically closed field of characteristic 13, the only CA polynomial

$$h = X^{20} + aX^{16} + cX^3 + dX$$

is X^{20} . Coefficients are allowed to vanish.

If $a = 0, c \neq 0$, normalize an H_3 common root to 1. Then $c = 4, d = 8$. A common H_1 root $w \neq 0$ satisfies

$$w^{19} + 4w^2 + 8 = 7w^{19} + 12w^2 + 8 = 0,$$

so $w^2 = 10, w^{19} = 4$. Since $10^9 = -1$, this forces $w = 9$, inconsistent with $9^2 = 3$. If $a = c = 0, d \neq 0$, the root and first-derivative equations give $-d = w^{19}$ and $-d = 7w^{19}$, also impossible.

If $a \neq 0$, normalize an H_{16} common root to 1, giving $a = 4$. Choose a nonzero H_3 common root v . This is forced when $c \neq 0$; when $c = 0$, $v = 1$ is available because $H_3 h(1) = 9 + 4 = 0$. The equations yield

$$c = 4v^{17} - 4v^{13}, \quad d = -5v^{19}, \quad F(v) = 5v^{19} - 4v^{17} + 4v^{13} - 5 = 0.$$

Thus $v, d \neq 0$. For an H_1 common root $w \neq 0$, subtracting $h'(w)$ from $3h(w)/w$ gives $w^{19} = 4v^{19}$. With $t = w/v$, $T = v^4$,

$$M(t) = t^{19} - 4 = 0, \quad (4t^2 - 1)T = 4(t^2 - t^{15}).$$

The zeros 6 and 7 of $D(t) = 4t^2 - 1$ have nineteenth powers 7 and 6, so $D \neq 0$. Put $B(t) = 4(t^2 - t^{15})$, so $T = B/D$. Substituting $v^4 = T$ in $F(v) = 0$ gives

$$5T^4 v^3 + 4T^3(1 - T)v = 5.$$

Squaring, using $v^4 = T$, and squaring once more yields the necessary equation

$$R(T) = (T^9 - T^8 - 1)^2 - T^{13}(-T^3 + 3T^2 - 6T + 3)^2 = 0.$$

No factor in this expression has been divided out. The remainder of $D^{19}R(B/D)$ modulo M is

$$\begin{aligned} H = 5t^{18} - 6t^{17} - 3t^{16} + 3t^{15} - 2t^{14} - 6t^{13} + t^{11} + 3t^{10} \\ - t^8 + t^7 - t^6 + 3t^5 + 6t^4 + 5t^3 - 4t^2 + 6. \end{aligned} \tag{S6}$$

But $M = (t - 4)Q$, where $Q = \sum_{j=0}^{18} 4^j t^{18-j}$ is irreducible: its roots are 4 times the primitive nineteenth roots of unity, and $\text{ord}_{19}(13) = 18$. The latter order follows from $13^6 = 11$ and $13^9 = -1 \pmod{19}$. Now $H(4) = 8$, and $H \neq 5Q$, since their t^{16} coefficients are 10 and 2. Thus M, H are coprime, a contradiction.

At 13 the invisible deficiency indices are 8 through 12. Lemma S.3 and S.1 therefore exclude the entire characteristic-zero closed support mask

$$T_D = \{4, 8, 9, 10, 11, 12, 17, 19\}. \tag{S7}$$

This removes D and the size-four survivor, proving already that a nontrivial centered polynomial needs at least six terms. Notice that this implication does not require every allowed coefficient to be nonzero.

3.4 S.4. The characteristic-13 lemma excluding E

Lemma S.4. The only characteristic-13 CA polynomial

$$h = X^{20} + aX^4 + cX^3 + dX$$

is X^{20} .

The case $a = 0$ is the first boundary argument of S.3. If $a \neq 0$, normalize an H_4 common root to 1; then $a = 4, d = -5 - c$. If $c = 0$, the first-derivative common-root equations imply $w^3 = 9, w^{19} = 8$, hence $w = 8$, inconsistent with $8^3 = 5$.

Choose an H_3 witness $v \neq 0$, and put $T = v^{16}$. The equations give

$$c = v(4T - 3), \quad d = -v^3(5T + 1), \quad Q(v, T) = (5T + 1)v^3 + (3 - 4T)v - 5 = 0.$$

If $d = 0$, then $T = 5, c = 8, v = 2$, inconsistent with $2^{16} = 3$. Thus a first-derivative witness w is nonzero. Set $t = w/v$. Subtracting seven times $h(w)/w$ from $h'(w)$ and then using the root equation gives

$$D(t)T = B(t), \quad D = 10t^2 + 4, \quad B = -t^3 + t^2 - 6,$$

$$U(t) = B(t)(t^{19} + 4t^2 - 5) + D(t)(4t^3 - 3t^2 - 1) = 0.$$

At the two zeros 6 and 7 of D , the values of B are 9 and 12, so $D \neq 0$. Explicitly,

$$U = -t^{22} + t^{21} - 6t^{19} - 3t^5 - 5t^3 + t^2.$$

Define

$$R(T) = \text{Res}_v(v^{16} - T, Q(v, T)), \quad H(t) = D^{19}R(B/D) \bmod U.$$

Here and below arrays list coefficients in ascending degree in $\mathbf{F}_{13}$. The complete small certificate is

$$\begin{aligned} R &= [1, 11, 0, 11, 9, 1, 5, 9, 0, 10, 8, 4, 11, 10, 4, 0, 12, 4, 8, 12] \\ H &= [6, 0, 10, 0, 3, 4, 1, 9, 2, 9, 7, 12, 9, 1, 8, 4, 4, 3, 11, 2, 10, 5] \\ CU &= [2, 4, 0, 1, 6, 9, 7, 10, 2, 10, 8, 8, 0, 5, 2, 7, 0, 7, 2, 1, 4] \\ CH &= [11, 0, 3, 1, 8, 7, 9, 1, 9, 8, 2, 0, 2, 7, 7, 6, 4, 9, 0, 10, 3, 6]. \end{aligned}$$

Direct multiplication gives $C_U U + C_H H = 1$. A putative solution has $U = H = 0$, contradiction. The resultant is independently checked as the determinant of the full 19×19 Sylvester matrix, rather than by the producer's reduced 5×5 determinant.

Valuation reduction now excludes the closed characteristic-zero deficiency mask

$$T_E = \{8, 9, 10, 11, 12, 16, 17, 19\},$$

and hence family E.

3.5 S.5. The larger characteristic-13 exclusion for B

Lemma S.5. The only characteristic-13 CA polynomial

$$h = X^{20} + aX^{17} + bX^4 + cX^3 + dX \tag{S8}$$

is X^{20} .

The chart $a = 0$ is Lemma S.4. If $a \neq 0$, normalize an H_{17} common root to 1, so $a = 4, d = -5 - b - c$. The remaining active Hasse derivatives are

$$H_4 h = 9X^{16} + 4X^{13} + b, \quad H_3 h = 9X^{17} + 3X^{14} + 4bX + c,$$

$$H_1 h = 7X^{19} + 3X^{16} + 4bX^3 + 3cX^2 + d.$$

For the whole chart $b = 0$, choose arbitrary witnesses v, w for orders 3 and 1 and substitute $c = -9v^{17} - 3v^{14}, d = -5 - c$. In $\mathbf{F}_{13}[v, w]$, define

$$F_3 = v(5v^{19} + v^{16} + d), \quad F_0 = w(w^{19} + 4w^{16} + cw^2 + d), \quad F_1 = 7w^{19} + 3w^{16} + 3cw^2 + d.$$

The supplemental sparse polynomials L_3, L_0, L_1 , with respectively 680, 694, and 712 terms, satisfy

$$L_3 F_3 + L_0 F_0 + L_1 F_1 = 1. \tag{S9}$$

This is ordinary ideal membership, with no saturation and no assumption $c, d, v, w \neq 0$.

For $b \neq 0$, an H_4 witness u is nonzero and

$$b = 4u^{13}(u^3 - 1), \quad c(u^2 - 1) = 5 + b - 5u^{19}.$$

The case $u = 1$ would give $b = 0$; at $u = -1$ the second equation is inconsistent. Thus put

$$q = u + 1, \quad C = \frac{5 + b - 5u^{19}}{u - 1}, \quad D = q(-5 - b) - C.$$

The quotient defining C is a polynomial, of degree 18. The actual coefficients are $c = C/q, d = D/q$, and $q \neq 0$. Define

$$\begin{aligned} P &= q(X^{19} + 4X^{16} + bX^3) + CX^2 + D, \\ J_3 &= q(9X^{17} + 3X^{14} + 4bX) + C, \quad J_1 = q(7X^{19} + 3X^{16} + 4bX^3) + 3CX^2 + D, \end{aligned}$$

and $R_i = \text{Res}_X(P, J_i)$, $i = 3, 1$. A zero common root for order 3 forces $C = 0$; a nonzero one forces $R_3 = 0$. Therefore the necessary equation is $CR_3 = 0$, and similarly $DR_1 = 0$. The supplemental univariate multipliers satisfy

$$A(u)C(u)R_3(u) + B(u)D(u)R_1(u) = (u+1)^{17}. \quad (\text{S10})$$

Its left side vanishes at a solution and its right side does not. This proves the lemma.

The saved resultant degrees are 359 and 395. The coefficient degrees of P, J_3, J_1 in u are at most 18, so the Sylvester bounds are 648 and 684. Independent exact evaluation at 685 distinct elements of $\mathbf{F}_{13^3}$, excluding $u = -1$ to preserve leading degrees, certifies the entire two resultant polynomials. Multiplication then checks (S9) and (S10) coefficient by coefficient. This is an identity check over an extension field with a proved degree bound, not a search for the absence of rational points.

The resulting characteristic-zero closed mask is

$$T_B = \{3, 8, 9, 10, 11, 12, 16, 17, 19\}.$$

It contains family B.

3.6 S.6. Family A: a nonempty reduction that forces exact collisions

Family A needs a different argument because its reduced coefficient family does contain a nontrivial CA polynomial.

Consider

$$h = X^{20} + aX^{17} + bX^{16} + cX^2 + dX$$

in characteristic 13. Its active derivatives are

$$H_{17} = 9X^3 + a, \quad H_{16} = 9X^4 + 4aX + b,$$

$$H_2 = 8X^{18} + 6aX^{15} + 3bX^{14} + c, \quad H_1 = 7X^{19} + 4aX^{16} + 3bX^{15} + 2cX + d.$$

The following complete algebraic classification includes every degeneration.

If $a = 0, b \neq 0$, normalize an H_{16} witness to 1: $b = 4, d = -5 - c$. With an H_2 witness v , substitute $c = -8v^{18} - 12v^{14}$. The supplied identity expresses 1 in the ordinary ideal

$$(h(v), h(w), 7w^{19} + 12w^{15} + 2cw + d) \subset \mathbf{F}_{13}[v, w]. \quad (\text{S11})$$

If $a = b = 0, c \neq 0$, normalization gives $c = 5, d = 7$. Then $H_1h - 7h/X = X + 10$, so its common nonzero root would be 3, whereas $(h/X)(3) = 12$. The remaining binomial $X^{20} + dX$, $d \neq 0$, is excluded as in S.3.

Thus $a \neq 0$. Normalize an H_{17} witness to 1, giving $a = 4, d = -5 - b - c$, and put $P = h/X$. The conditions for orders 2 and 1 imply

$$c \text{ Res}_X(P, H_2h) = 0, \quad d \text{ Res}_X(P, H_1h) = 0. \quad (\text{S12})$$

Multiplication by c, d retains zero-coefficient charts. There are three univariate cases:

Case	Parameter and coefficient substitution	Gcd of the two polynomials in (S12)
$b = 0$	$c, d = -5 - c$	1
H_{16} -witness $u = 1$	$b = 1, d = -6 - c$	$c - 4$
$b \neq 0, u \neq 1$	$b = 4u^4 - 3u, c = (5 + b - 5u^{19} - u^{16})/(u - 1), d = -5 - b - c$	1

The last numerator is divisible by $u - 1$; it is a degree-18 polynomial after division. The witnesses there also satisfy $u \neq 0$, since $b \neq 0$. The supplied resultants are exact Sylvester determinants; the checked parameter-degree bounds for the multiplied resultants are 38, 39 in the first two cases and 684, 702 in the last. Evaluation at more than each bound over $\mathbf{F}_{13^3}$, followed by exact polynomial gcd, proves the table.

Consequently the unique normalized nonmonomial seed is

$$h_A = X^{20} + 4X^{17} + X^{16} + 4X^2 + 3X.$$

Exact gcds are

$$\gcd(h_A, H_{17}h_A) = \gcd(h_A, H_{16}h_A) = \gcd(h_A, H_2h_A) = X - 1,$$

and $h'_A(1) = 11 \neq 0$.

Now take a characteristic-zero polynomial whose support is contained in A 's exponent mask. Apply S.1 at 13; the X^{10} coefficient disappears, and the preceding classification shows that the X^{17} coefficient is a unit. Its H_{17} witness is therefore a unit and may be scaled to 1 while preserving integrality. The other two witnesses reduce to the simple root 1. They equal 1 exactly: the polynomial divided difference

$$\frac{F(r) - F(1)}{r - 1}$$

is integral and reduces to $h'_A(1)$, so is a unit; its product with $r - 1$ is zero. This proves exact equality without a completeness or unramified-lifting assumption.

Writing t for the ordinary X^{10} coefficient, the exact equations at 1 give

$$F = X^{20} - 1140X^{17} + 14535X^{16} + tX^{10} + (-1589350 - 45t)X^2 + (1575954 + 44t)X. \quad (\text{S13})$$

Let $Q = F/X$, and define the exact integer polynomials

$$R_{10}(t) = \text{Res}_X(Q, H_{10}F), \quad R_1(t) = \text{Res}_X(Q, H_1F).$$

Their degrees are 19 and 28; reduction modulo 101 preserves both degrees and gives gcd 1. Hence they are coprime over $\mathbf{Q}$. For $t \neq 0$, an H_{10} witness is nonzero, forcing $R_{10}(t) = 0$. The H_1 condition always forces $R_1(t) = 0$: if its witness is zero, the linear coefficient vanishes and zero is then also a root of Q . This contradicts coprimality. For $t = 0$, the separately checked value $R_1(0) \neq 0$ supplies the contradiction.

The integer resultants are certified at 30 and 39 distinct integer parameter values by exact Sylvester determinants: the a priori parameter-degree bounds are 29 and 38. This certifies the resultant identities before the degree-preserving modular gcd check. Family A is thus excluded as a closed support, including the $t = 0$ boundary.

3.7 S.7. Family C: reduction, all six residue rows, and ramified precision

It remains to exclude exact support C:

$$F = X^{20} + AX^{16} + BX^{15} + CX^{10} + DX^3 + EX, \quad ABCDE \neq 0. \quad (\text{S14})$$

Normalize integrally at 13. An H_{10} witness gives

$$184756z^{10} + 8008Az^6 + 3003Bz^5 + C = 0.$$

The first three numerical coefficients are divisible by 13; hence $C \in 13O$. The reduced shape is $h = X^{20} + aX^{16} + bX^{15} + dX^3 + eX$.

We must justify normalizing an H_{16} witness. If $a = e = 0$ in a nonmonomial seed, $b = 0$ gives an immediate root/ H_3 contradiction. For $b \neq 0$, normalize an H_{15} witness to 1; then $b = 5, d = 7$, and an H_3 witness would satisfy $v^{17} = 5, v^5 = -1$. These imply $v^2 = 8, v = 1$, a contradiction. Thus $a = 0$ would require $e \neq 0$. But $\overline{H_{16}F} = 9X^4$ would force its witness into the simple residue root zero. Divided-difference uniqueness would make that exact witness zero, contrary to $A \neq 0$. Therefore A is a unit. Normalize its witness to 1:

$$A = -4845, \quad F(1) = 0, \quad E = -1 - A - B - C - D. \quad (\text{S15})$$

3.7.1 S.7.1. Exact algebraic classification of the reduced seeds

Write the seed now as $h = X^{20} + 4X^{16} + bX^{15} + cX^3 + dX$. Lemma S.3 excludes $b = 0$, so an H_{15} witness u is nonzero. At $u = -1$ the equations give $h(-1) = 10$; treat $u = 1$ separately. For $u \neq 0, \pm 1$,

$$b = 5u^5 + u, \quad N = 5 + b - 6u^{19} - 5u^{15}, \quad C_n = N/(u-1), \quad q = u+1, \quad D_n = q(8-b) - C_n, \\ c = C_n/q, \quad d = D_n/q.$$

Again C_n is a polynomial of degree 18. Set

$$P = q(X^{19} + 4X^{15} + bX^{14}) + C_nX^2 + D_n, \\ J_3 = q(9X^{17} + 4X^{13}) + C_n, \quad J_1 = q(7X^{19} + 12X^{15} + 2bX^{14}) + 3C_nX^2 + D_n,$$

and $R_i = \text{Res}_X(P, J_i)$. The necessary conditions are $C_nR_3 = D_nR_1 = 0$, including coefficient loss. The exact supplemental identity is

$$UR_3C_n + VR_1D_n = (u+1)^{17}(u-1)(u-2)(u^2+4u-2). \quad (\text{S16})$$

It leaves $u = 2$, giving $(b, c, d) = (6, 3, 12)$, or $u^2 + 4u - 2 = 0$, giving $(6, 2, 0)$.

For $u = 1$, $b = 6, d = 2 - c$. Use $P = X^{19} + 4X^{15} + 6X^{14} + cX^2 + 2 - c$ and the corresponding unscaled derivatives

$$J_3 = 9X^{17} + 4X^{13} + c, \quad J_1 = 7X^{19} + 12X^{15} + 12X^{14} + 3cX^2 + 2 - c.$$

A second identity has left side $UcR_3 + V(2-c)R_1$ and right side

$$(c-2)(c-3)(c-10). \quad (\text{S17})$$

Thus the complete list is $(6, 3, 12), (6, 2, 0), (6, 10, 5)$. The generic resultants have degree bounds 648 and 684 and are certified at 685 extension-field points; the special bounds are 36 and 38 and require 39 points. These identities prove completeness over every algebraically closed characteristic-13 field; there is no restriction to base-field coefficients or witnesses.

For the seed $(6, 2, 0)$, $\gcd(h, h') = X^2$. In characteristic zero, an H_1 witness $w \neq 0$ must therefore have positive valuation; it cannot be zero because $E \neq 0$. Yet

$$F'(w) - F(w)/w = 19w^{19} + 15Aw^{15} + 14Bw^{14} + 9Cw^9 + 2Dw^2 = 0$$

has its last term of uniquely least valuation, as D is a unit. This excludes that entire coefficient point, including its extension-field marked witnesses.

Put

$$h_3 = X^{20} + 4X^{16} + 6X^{15} + 3X^3 + 12X, \quad h_{10} = X^{20} + 4X^{16} + 6X^{15} + 10X^3 + 5X.$$

The exact monic gcds that determine the marked roots are

	H_{15}	H_3	H_1
h_3	$(X-1)^2(X-2)$	$X-2$	$(X-1)(X-4)^2$
h_{10}	$X-1$	$X-11$	$(X-3)(X-11)$

The relevant multiplicities in h_3 are 1 at 0 and 2, 2 at 1, and 3 at 4; in h_{10} they are 1 at 0 and 1, and 2 at 3 and 11. These are checked by division and evaluation of Hasse derivatives.

Let u, v, w, z denote the common-root witnesses for Hasse orders 15, 3, 1, 10. The gcds give precisely the six rows below:

Row	Seed	$\bar{v}$	$\bar{u}$	$\bar{w}$	Exact consequences
1	h_3	2	1	1	$u = w = 1$

Row	Seed	$\bar{v}$	$\bar{u}$	$\bar{w}$	Exact consequences
2	h_3	2	1	4	none initially
3	h_3	2	2	1	$u = v, w = 1$
4	h_3	2	2	4	$u = v$
5	h_{10}	11	1	3	$u = 1$
6	h_{10}	11	1	11	$u = 1, v = w$

The exact consequences use only cluster multiplicity. A simple residue root has a unique exact root above it. A double residue cluster containing an exact repeated root is exhausted by that repeated root; if it also contains the already fixed root 1, the repeated root equals 1. Root counting here is valid because the monic polynomial factors into integral linear factors, whose reductions record the cluster multiplicities.

3.7.2 S.7.2. Two precision lemmas

Normalize $\nu(13) = 1$. Suppose integral polynomial equations have a zero x congruent to an integral base point b , their Jacobian at b is invertible over O , and their defects at b and any external parameter errors have valuation at least $\lambda > 0$. Then every coordinate of $x - b$ has valuation at least λ . Indeed, if their minimum γ were $0 < \gamma < \lambda$, the linear term would have minimum valuation γ , preserved by an invertible integral matrix. Constant errors and all terms quadratic in the deviations have larger valuation. The equations cannot vanish. We call this the **unit-Jacobian bound**.

A second observation is needed at the triple residue root 4 of h_3 . Once the coefficients agree with an integer lift modulo $13O$, $F'(4), H_2F(4) \in 13O$, while $3H_3F(4)$ is a unit. If $w \equiv 4$ and $F'(w) = 0$, Taylor expansion forces $\nu(w - 4) \geq 1/2$; otherwise its quadratic term has uniquely least valuation. If also $F(w) = 0$, Taylor expansion of F then shows $\nu(F(4)) > 1$. Consequently $F(4)/13 = 0$. This does **not** claim $w - 4 \in 13O$.

3.7.3 S.7.3. The exact $u = 1$ cases

The order-15 equation gives $B = B_0 = 62016$. At $(v, D) = (2, 3)$ and $(11, 10)$, the Jacobians of $(F(v), H_3F(v))$ in (v, D) are respectively

$$\begin{pmatrix} 9 & 6 \\ 4 & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 7 \\ 4 & 1 \end{pmatrix} \pmod{13},$$

both of determinant 11. Since $C \in 13O$, the bound proves

$$v = v_0 + 13t, \quad D = d_0 + 13l, \quad C = 13k, \quad t, l, k \in O.$$

Dividing the exact equations by 13 gives, for $(v_0, d_0) = (2, 3)$,

$$9t + 6l + 8k + 12 = 0, \quad 4t + l + 7k + 6 = 0; \tag{S18}$$

for $(11, 10)$, it gives

$$7l + 12k + 2 = 0, \quad 4t + l + 6k + 3 = 0. \tag{S19}$$

All equations in the following table are residue equations.

$(\bar{v}, \bar{w})$	Additional equation and reason	$(\bar{t}, \bar{l}, \bar{k})$
(2, 1)	$2l + 9k + 7 = 0$, from $F'(1) = 0$	(1, 3, 0)
(2, 4)	$8l + 5k + 1 = 0$, by the triple-cluster bound	(5, 7, 12)
(11, 3)	$11l + 9 = 0$, from $F(w) = 0$	(8, 11, 1)
(11, 11)	$t + 11l + k + 7 = 0$, from $F'(v) = 0$	(0, 6, 5)

For the third row, h'_{10} has a simple root at 3, so first apply the one-variable bound to obtain $w - 3 \in 13O$; its displacement contributes zero to $F(w)/13$ because 3 is a double root of h_{10} . In the fourth row, $v = w$ is an exact collision.

The integral monic middle derivative $H_{10}F/184756$ reduces to

$$g_\lambda = X^{10} + 11X^6 + 7X^5 + \lambda, \quad \lambda = 9\bar{k}.$$

Exact Euclidean gcds are

$$\gcd(h_3, g_0) = X, \quad \gcd(h_3, g_4) = 1, \quad \gcd(h_{10}, g_9) = \gcd(h_{10}, g_6) = 1. \quad (\text{S20})$$

The last three cases are impossible. In the first, the middle witness reduces to the simple root zero and hence is exactly zero; this would give $C = H_{10}F(0) = 0$, contrary to exact support. This excludes rows 1, 5, 6 and the $u = 1$ part of row 2.

3.7.4 S.7.4. Row 2 without an assumed exact collision

Here the order-15 equation gives $B(u) = 77520u - 15504u^5$. For $s = u - 1$,

$$B(u) - B_0 = -15504s^2(10 + 10s + 5s^2 + s^3). \quad (\text{S21})$$

If $r = \nu(s) > 0$, this has valuation $2r$. The same two-variable bound now gives

$$\min\{\nu(v-2), \nu(D-3)\} \geq \min\{1, 2r\}.$$

Moreover

$$F'(1) = 13 \cdot 61198 + 14(B - B_0) + 9C + 2(D - 3), \quad H_2F(1) \equiv 9.$$

If $r < 1$, the exact divided-difference equation

$$0 = \frac{F(u) - F(1)}{u - 1} = F'(1) + H_2F(1)s + H_3F(1)s^2 + \dots$$

has a uniquely smallest term $H_2F(1)s$: the first term has valuation at least $\min\{1, 2r\} > r$, and the remaining terms have valuation at least $2r$. Thus $r \geq 1$. It follows that $B - B_0 \in 13^2O$ and $v - 2, D - 3 \in 13O$. These conclusions also hold if $s = 0$. The $(2, 4)$ first-jet calculation in S.7.3 is unchanged and again gives $\gcd(h_3, g_4) = 1$, excluding all of row 2.

3.7.5 S.7.5. Rows 3 and 4: the collision $u = v$

Substitute $B(u) = 77520u - 15504u^5$ in $F(u) = H_3F(u) = 0$. The Jacobian in (u, D) at $(2, 3)$, including differentiation of $B(u)$, is

$$\begin{pmatrix} 10 & 6 \\ 4 & 1 \end{pmatrix} \pmod{13}, \quad \det = 12.$$

Hence $u = 2 + 13r, D = 3 + 13l, C = 13k$, with integral r, l, k . The first two divided equations reduce to

$$10r + 6l + 8k + 7 = 0, \quad 4r + l + 7k + 6 = 0. \quad (\text{S22})$$

In row 4, the triple-cluster bound supplies $10r + 8l + 5k + 3 = 0$. Together they give $(\bar{r}, \bar{l}, \bar{k}) = (5, 7, 12)$, so (S20) excludes the row.

In row 3, $w = 1$, and $F'(1) = 0$ gives $11r + 2l + 9k + 4 = 0$. The solution is $(12, 3, 3)$. Here

$$\gcd(h_3, g_1) = X - 4, \quad g'_1(4) = 3,$$

so the middle witness z reduces to 4, but a further precision argument is essential. After substitution, the three exact expressions $F(u), H_3F(u), F'(1)$ are coefficientwise divisible by 13 in $\mathbf{Z}[r, l, k]$. Their divided polynomials reduce to the three affine equations above, whose coefficient matrix has determinant 3. Apply the bound again at $(12, 3, 3)$, obtaining

$$r - 12, l - 3, k - 3 \in 13O. \quad (\text{S23})$$

It follows that the monic middle derivative evaluated at 4 lies in $13O$; its derivative there is a unit. The one-variable bound gives $z - 4 \in 13O$. But the exact expansion is

$$\overline{F(4)/13} = 10\bar{r} + 8\bar{l} + 5\bar{k} + 3 = 6.$$

Since $F'(4), H_2F(4) \in 13O$, Taylor expansion implies $F(z) - F(4) \in 13^2O$, contradicting $F(z) = 0$. This excludes row 3, completing the proof for C and hence Theorem S.

3.8 S.8. Exact certificate dependencies and their scope

For the accompanying evidence tree, let E denote `evidence/seven-terms`. Within E , let P denote `dependencies/casas-alvero-sixterm`. Within P , let Q denote `dependencies/casas-alvero-structural` and R denote `dependencies/casas-alvero-extension`. These aliases locate coefficient data; they do not introduce additional proof assumptions.

Mathematical item	Exact data / replay, relative to the indicated directory
Support sieve (S2)–(S4), five A–E supports	Q : <code>sixterm/old-baseline-and-groups.json</code> ; <code>sixterm/old_baseline_and_groups.py</code> ; <code>sixterm/enumerate_sixterm.py</code> ; <code>sixterm/apply_two_visible.py</code>
Lemma S.3 remainder (S6)	R : <code>check_mod13_explanation.py</code>
Lemma S.4 resultant and displayed Bézout identity	Q : <code>sixterm/last-mask-probe.json</code> ; <code>sixterm/verify_last_mask.py</code>
Lemma S.5 identities (S9), (S10) and resultant arrays	P : <code>B/certificate.json</code> ; <code>B/verify_certificate.py</code>
A boundary ideal identity (S11)	P : <code>A/seed13-a-zero-certificate.txt</code> ; <code>A/check_seed_certificates.py</code> zero
A univariate classification (S12)	P : <code>A/seed13-univariate.txt</code> ; <code>A/check_seed_univariate.py</code>
A characteristic-zero resultants of (S13)	P : <code>A/collision-resultants.log</code> ; <code>A/check_collision_resultants.py</code>
C classification identities (S16), (S17)	P : <code>C/univariate-certificate.json</code> ; <code>C/verify_univariate.py</code>
C exact jets and gcds (S18)–(S23)	E : <code>u_one/check_jets.py</code> ; <code>cluster/check_jets.py</code> ; <code>u_equals_v/check_jets.py</code>
Separate C arithmetic reconstructions	E : <code>u_one/check_u_equals_v_audit.py</code> ; <code>structural_audit/check_independent_jets.py</code>
Complete package and six-row coverage replay	E : <code>replay.py</code> ; <code>case_coverage.json</code> ; <code>MANIFEST.json</code>

The B certificate has SHA-256 981ff911b8493e9a3cb37dea620dd63249e6bd2bff9c93f16aec2670e88179c7. The package manifest fixes the other certificate bytes. The B and C resultant replays share a finite-field arithmetic implementation; their independence is from the Singular producer, not from each other. The C jet computations have additional separately written reconstructions.

The theorem depends on the written normalization, denominator, degeneration, root-cluster, and precision arguments, as well as the finite identities. A PASS receipt alone does not establish those arguments. Conversely, the elementary implications do not excuse an unchecked resultant identity. The optional large characteristic-zero resultant proof of C, exploratory Gröbner outputs, and older modular finite-module certificates are not dependencies of this proof.

This is a proof dependency consolidation of an existing result. It makes no assertion of historical novelty, publication significance, external refereeing, proof-assistant verification, or a solution in unrestricted degree 20.

3.9 S.9. The fourteen-support frontier before the additional exclusions

The frozen unrestricted support inventory contains the following fourteen seven-total-term supports. They are copied from `evidence/full/support_frontier/inventory.json`; no new enumeration is used here:

$$\begin{array}{lll}
\{2, 3, 4, 10, 12, 19\}, & \{3, 4, 9, 10, 12, 19\}, & \{3, 4, 5, 10, 13, 19\}, \\
\{3, 4, 10, 12, 15, 19\}, & \{3, 7, 9, 10, 16, 19\}, & \{3, 6, 10, 16, 17, 19\}, \\
\{7, 8, 10, 16, 17, 19\}, & \{10, 12, 13, 16, 17, 19\}, & \{6, 10, 15, 16, 17, 19\}, \\
\{9, 10, 15, 16, 17, 19\}, & \{2, 4, 10, 12, 18, 19\}, & \{3, 4, 10, 13, 18, 19\}, \\
\{2, 4, 10, 17, 18, 19\}, & \{4, 10, 16, 17, 18, 19\}. &
\end{array}$$

None is eliminated by Theorem S’s completed A–E argument: adding an extra coefficient changes both the reduced seed and the exact collision equations. The characteristic-17 and cross-prime analyses elsewhere in

the report give necessary restrictions or exclusions for specified residue branches. Those statements must retain their residue and normalization hypotheses; a local branch exclusion is not, by itself, an exclusion of an entire support in this list. The following sections supply the missing global coverage bridge for six of these supports, leaving eight. Neither the A–E argument nor those additional exclusions supplies an eight-term theorem. The remaining coefficient and witness branches, including ramified lifts, are unresolved.

4 Global exclusion of the centered support $\{2,4,10,12,18,19\}$

24 September 2026. This corollary assembles the existing complete characteristic-17 seed classification and the existing row-4 support exclusion. It introduces no new census or lifting computation.

Corollary. No nontrivial characteristic-zero degree-20 Casas–Alvero polynomial has exact centered deficiency support

$$S = \{2, 4, 10, 12, 18, 19\}.$$

Equivalently, after centering the mean root at zero and making the polynomial monic, there is no CA polynomial of the form

$$X^{20} + c_2X^{18} + c_4X^{16} + c_{10}X^{10} + c_{12}X^8 + c_{18}X^2 + c_{19}X, \quad \prod_{j \in S} c_j \neq 0.$$

The leading term is not counted in the deficiency set: this is one exact seven-monomial support. The assertion is not conditional on already belonging to row 4.

4.1 Proof

Write

$$f(X) = \sum_{j=0}^{20} \binom{20}{j} a_j X^{20-j}, \quad a_0 = 1, \quad a_1 = a_{20} = 0.$$

The condition $a_1 = 0$ is centering. The CA condition for the nineteenth derivative makes its root zero a root of f , giving $a_{20} = 0$. Ordinary and binomial-normalized coefficients have the same exact zero pattern in characteristic zero.

First reduce to algebraic coefficients. The coefficients and nineteen chosen common-root witnesses satisfy polynomial equations over $\mathbb{Q}$. Add the equations $a_j = 0$ for every $j \notin S$, apart from $a_0 = 1$, and

$$v \prod_{j \in S} a_j = 1.$$

A solution over any characteristic-zero extension makes this ideal proper; the weak Nullstellensatz supplies a point over $\mathbb{Q}$. This retains exact support, not only containment. It therefore suffices to exclude algebraic points.

Choose a valuation above 17, extend to a field containing all roots, and divide the variable by a nonzero root of minimum valuation: explicitly replace $f(X)$ by $f(\lambda X)/\lambda^{20}$. All roots are now integral and one root is exactly one. This operation sends a_j to a_j/λ^j , so it preserves every exact zero and nonzero coefficient. No translation is made after centering.

The normalized monic derivatives are

$$G_j(X) = \frac{H_{20-j}f(X)}{\binom{20}{j}} = \sum_{i=0}^j \binom{j}{i} a_i X^{j-i}.$$

Every G_j has a common root w_j with f ; the witness w_j is integral. Induction in j , using the coefficient one on a_j , proves a_j integral. This step is necessary because ordinary coefficient integrality alone would not justify Lucas visibility.

For $4 \leq j \leq 16$, the integer $\binom{20}{j}$ is divisible by 17. All other nonleading coefficients absent from S vanish exactly. Thus the reduction has the visible form

$$h = X^{20} + aX^{18} + bX^{17} + cX^3 + dX^2 + eX \quad \text{with } b = c = 0.$$

Reducing the ordinary Hasse common-root equations proves that h is Hasse-CA. In particular the visible derivatives retain their common-root identities; the Hasse orders 4 through 16 vanish identically on the visible model. The reduction is not X^{20} , because the retained exact root one reduces to a nonzero root.

The complete algebraic-closure classification has nine nonmonomial rows up to nonzero variable scaling. Only row 4 has both $b = c = 0$. This remains true if a_2, a_{18} , or a_{19} initially has positive valuation; none of those coefficient residues was assumed nonzero. Indeed, the coefficient-zero boundary charts are included in the classification, and their only additional all-zero tuple is the already excluded monomial.

Lift a classification scaling to a unit, extending the local field if necessary. The resulting reduction is

$$h = X^{20} - 3X^{18} + 11X^2 + 8X.$$

Centering and exact support are still unchanged. At this stage $\bar{a}_2 = -1$. Choose an actual common root z of f and $G_2 = X^2 + a_2$. Its reduction is 1 or -1 . In the row-4 seed,

$$h(1) = 0, \quad h(-1) = 1.$$

Therefore z reduces to one and is a unit. Scaling by this actual witness leaves the residue seed unchanged and gives

$$a_2 = -1, \quad a_3 = a_{17} = 0$$

exactly. The active middle indices are precisely $J = \{4, 10, 12\}$.

This is the exact family excluded by the existing row-4 smallest-support theorem. Its complete residue census retains all seventeen nonzero witness residues, both critical-root orientations, and active coefficients whose residues vanish. Its two residue survivors have simple selected witness classes and a unit lifting Jacobian. The final critical-value certificate is

$$f(R) = 17^2(9 + 4\alpha) \pmod{17^3}, \quad \alpha^2 + 3\alpha + 3 = 0,$$

which is nonzero. The theorem's analytic division and uniqueness arguments permit arbitrary ramified valued extensions. Thus the normalization above falls within its proved scope and is impossible. This excludes the algebraic specialization and hence the original characteristic-zero point. $\square$

4.2 Exact dependencies and verification boundary

1. **Complete seed classification**: classification over the whole algebraic closure, with coefficient-zero charts and monomial retained separately. Its certificate checker and **independent mathematical audit** validate completeness; checking only the nine displayed examples would not suffice.
2. **Normalization and integrality**: the recurrence proof is reproduced above. The general simple-mean argument is not an extra bridge needed to select row 4, whose mean residue root is already simple.
3. **Row-4 smallest-support exclusion**: the local computational theorem for $J = \{4, 10, 12\}$, together with its **separate audit** and standard-library checker.

The corollary's normalization and support-preservation bridge received a separate bounded algebraic audit. No existing evidence files were modified and no large computation was rerun.

This does not exclude the whole row-4 branch, every seven-term support, or degree 20. It does not assert the same conclusion merely from support containment, and makes no claim about priority or an assessment grade.

5 Six global exclusions in the seven-term support frontier

24 September 2026. This note independently verifies a bounded new use of the existing ramification-safe row-8 divided identity. The computation contains 1,216 residue assignments in total. It is not the unrestricted row-8 census.

5.1 Statement

Proposition. The characteristic-17 row-8 branch cannot occur for any of the seven exact centered deficiency supports in the table below.

Exact deficiency support	Active middle indices J	Assignments $4^{ J }$	After divided identity	Possible seed rows before this test
$\{2, 3, 4, 10, 12, 19\}$	$4, 10, 12$	64	0	5,8
$\{3, 4, 9, 10, 12, 19\}$	$4, 9, 10, 12$	256	0	8
$\{3, 4, 5, 10, 13, 19\}$	$4, 5, 10, 13$	256	0	8
$\{3, 4, 10, 12, 15, 19\}$	$4, 10, 12, 15$	256	0	8
$\{3, 7, 9, 10, 16, 19\}$	$7, 9, 10, 16$	256	0	8
$\{3, 6, 10, 16, 17, 19\}$	$6, 10, 16$	64	0	1,8
$\{3, 4, 10, 13, 18, 19\}$	$4, 10, 13$	64	0	8,9

No unit-root collision filter is used: every row is already empty after the divided identity alone.

Global corollary. The following five exact centered seven-term supports cannot occur for a characteristic-zero degree-20 CA polynomial:

$$\begin{aligned} &\{3, 4, 9, 10, 12, 19\}, & \{3, 4, 5, 10, 13, 19\}, \\ &\{3, 4, 10, 12, 15, 19\}, & \{3, 7, 9, 10, 16, 19\}, \\ && \{3, 4, 10, 13, 18, 19\}. \end{aligned}$$

The first four have no other possible seed. For the fifth, the previously proved complete row-9 exclusion with $J = \{4, 10, 13\}$ removes the other possible seed. The first and sixth rows of the table remain unresolved globally by this argument: rows 5 and 1, respectively, are still possible.

5.2 1. Normalization and the inherited necessary identity

Use the centered monic notation

$$f = \sum_{j=0}^{20} C_j a_j X^{20-j}, \quad C_j = \binom{20}{j}, \quad a_0 = 1, \quad a_1 = a_{20} = 0.$$

Integral root normalization at 17 and the normalized derivative recurrence make all roots and all a_j integral. Scaling, including the later unit-witness normalization, preserves exact support. Algebraic specialization retains an exact support by adjoining the inverse of the product of its active coefficients. These details are given explicitly in [GLOBAL_SUPPORT_COROLLARY.md](#).

In row 8,

$$\bar{f} = X^{17}(X^3 - 1).$$

The common witness for G_3 is a unit; scale it exactly to one. Then $f(1) = G_3(1) = 0$, so

$$a_3 = -1 - 3a_2. \tag{1}$$

The seed is unchanged by this additional scaling because the residue of the scaling witness is a cube root of unity.

The existing proof [ROW8_BOUND_AND_DIVIDED_IDENTITY.md](#) gives

$$\sum_{j=4}^{16} (C_j/17) \bar{a}_j = -1. \tag{2}$$

For completeness, its valuation bridge has no unramified-field assumption. Let $\delta > 0$ be the least valuation of a nonzero root in the seventeen-root zero cluster, normalized by $\nu(17) = 1$. The cluster has such a root

because the exact mean is simple; for the seven supports here this also follows immediately from the nonzero linear coefficient a_{19} . The witnesses for $G_2, G_{17}, G_{18}, G_{19}$ reduce to zero. Their exact equations imply

$$\begin{aligned}\nu(a_2) &\geq 2\delta, \\ \nu(a_{17}) &\geq \min(17\delta, 1 + \delta), \\ \nu(a_{18}) &\geq \min(18\delta, 1 + 2\delta), \\ \nu(a_{19}) &\geq \min(19\delta, 1 + 3\delta).\end{aligned}$$

At a root attaining δ , the X^{17} term has value 17δ ; every other term has value at least $\min(20\delta, 1 + 4\delta)$. Unique minimal valuation would prevent cancellation if $\delta < 1/13$. Thus $\delta \geq 1/13$, and

$$\nu(a_{17}) \geq 1 + \delta, \quad \nu(a_{18}) \geq 1 + 2\delta, \quad \nu(a_{19}) \geq 1 + 3\delta.$$

Substitution of (1) into $f(1) = 0$ gives the exact equation

$$0 = -17 \cdot 67 - 17 \cdot 190a_2 + \sum_{j=4}^{16} C_j a_j + 1140a_{17} + 190a_{18} + 20a_{19}.$$

Divide by 17 and reduce. The displayed strict bounds remove the last three terms, and $\nu(a_2) > 0$ removes the second. Since $67 = -1$ modulo 17, equation (2) follows. Fractional valuations are allowed throughout.

5.3 2. Canonical witnesses at inactive coefficients

Let S be an exact support from the table and $J = S \cap \{4, \dots, 16\}$. If $j \notin J$, then $a_j = 0$ exactly. Because $G_j(0) = a_j$ and $f(0) = 0$, zero is an exact common witness for this derivative. Choose it. These choices can be made independently: the CA condition requires the existence of a witness for each order and imposes no requirement that these particular choices differ.

For an active coefficient $j \in J$, every possible witness residue must be retained:

$$\rho_j \in \{0, 1, \zeta, \zeta^2\}, \quad \zeta^2 + \zeta + 1 = 0 \text{ in } \mathbb{F}_{17^2}.$$

In particular, an active coefficient or a nonzero exact witness may have zero residue. Neither is discarded.

The residue equations start with $\bar{a}_0 = 1, \bar{a}_1 = \bar{a}_2 = 0, \bar{a}_3 = -1$, and determine successively

$$\bar{a}_j = - \sum_{i=0}^{j-1} \binom{j}{i} \bar{a}_i \rho_j^{j-i}, \quad 4 \leq j \leq 16. \quad (3)$$

At every inactive index the chosen $\rho_j = 0$ makes this coefficient zero. Hence each of the $4^{|J|}$ active label assignments determines one complete residue coefficient vector. Every actual counterexample with that support gives at least one of these assignments.

The finite field restriction is a conclusion of (3), not an assumption about the ambient residue field or about the ramification of a lift. The visible seed has no roots outside this four-element domain in the algebraic closure. An empty necessary-condition census therefore excludes row 8 for the exact support.

5.4 3. Independent finite check

[check_row8_seven_term.py](#) imports no producer or existing checker. It represents the field as $\mathbb{F}_{17}[s]/(s^2-14)$, verifies that 14 is a nonsquare, and uses the complete domain

$$0, \quad 1, \quad 8 + 9s, \quad 8 + 8s.$$

The three nonzero entries are distinct cube roots of unity. This presentation differs from the earlier producer's $1, \zeta$ basis.

The checker traverses the Cartesian product independently for each support, implements the full recurrence (3), evaluates (2), and records a histogram of all attained sums. It verifies the expected assignment count and finds no sum equal to -1 in any of the seven cases. Normal and optimized Python runs produce identical receipts:

- [normal receipt](#);
- [optimized receipt](#).

The checker also binds the support list to the existing support inventory, verifies that these are exactly its seven size-six deficiency sets containing 3, and checks the seed routing in the table. It does not rerun the much larger predecessor inventory. The witness-canonicalization argument received a separate bounded algebraic review.

5.5 4. From branch exclusions to global statements

The complete nine-seed classification applies after normalization because a unit root is retained, excluding monomial reduction. Exact coefficient zeros remain zero under reduction and scaling. Here is the complete table, in ordinary visible coefficients $h = X^{20} + aX^{18} + bX^{17} + cX^3 + dX^2 + eX$:

Row	a	b	c	d	e
1	0	0	16	0	0
2	14	0	16	0	3
3	14	8	16	12	0
4	14	0	0	11	8
5	14	2	0	0	0
6	14	2	0	11	6
7	14	2	0	14	3
8	0	16	0	0	0
9	0	16	0	14	3

The table is an input from the independently checked [algebraic-closure classification](#) and its [audit](#). The new checker verifies routing against this table; it does not reprove the classification merely by checking the nine examples.

For each of the seven supports, the table retains exactly the seed rows whose nonzero visible coefficients do not occur at an absent index. This argument never infers an exact coefficient zero from a zero residue.

Four supports omit 2, 17, 18 and contain 3, 19. The visible polynomial can then have only its X^{17} and X coefficients nonzero below the leading term. Row 8 is the only compatible nonmonomial seed. Their row-8 exclusions are therefore global.

For $\{3, 4, 10, 13, 18, 19\}$, only rows 8 and 9 are compatible. The row-9 canonical middle support is $J = \{4, 10, 13\}$, which is already completely excluded in [BATCH1_PROOF.md](#). That prior computation covers its entire algebraic residue domain and arbitrary ramification. Combining it with the new row-8 exclusion proves the fifth global exclusion.

For $\{2, 3, 4, 10, 12, 19\}$, row 5 remains compatible. For $\{3, 6, 10, 16, 17, 19\}$, row 1 remains compatible. The present argument excludes neither remaining branch.

5.6 5. Exact coverage increment

The predecessor inventory contained fourteen possible exact seven-term supports. The five global exclusions above and the separate global row-4 corollary for $\{2, 4, 10, 12, 18, 19\}$ remove six distinct entries. The remaining eight are

$$\begin{aligned}
&\{2, 3, 4, 10, 12, 19\}, \quad \{3, 6, 10, 16, 17, 19\}, \\
&\{7, 8, 10, 16, 17, 19\}, \quad \{10, 12, 13, 16, 17, 19\}, \\
&\{6, 10, 15, 16, 17, 19\}, \quad \{9, 10, 15, 16, 17, 19\}, \\
&\{2, 4, 10, 17, 18, 19\}, \quad \{4, 10, 16, 17, 18, 19\}.
\end{aligned}$$

Explicitly, the six excluded exact supports are

$$\begin{aligned} &\{2, 4, 10, 12, 18, 19\}, \quad \{3, 4, 9, 10, 12, 19\}, \\ &\{3, 4, 5, 10, 13, 19\}, \quad \{3, 4, 10, 12, 15, 19\}, \\ &\{3, 7, 9, 10, 16, 19\}, \quad \{3, 4, 10, 13, 18, 19\}. \end{aligned}$$

The eight remaining sets are necessary possibilities, not asserted realizable supports. The coverage statement depends on the predecessor inventory and its underlying imported restrictions. It does not raise the seven-term lower bound to eight terms, exclude the whole row-8 branch, or solve degree 20. No priority or assessment-grade claim is made.

6 Marked splitting of a local resultant algebra

6.1 Input and notation

The input is the audited normalization and the explicit integer polynomials in [the explicit local presentation](#). In particular,

$$f_u = (X - 1)^2 g_u, \quad g_u(1) = 17T(u),$$

where g_u is monic of degree eighteen and

$$T = -8037 + \sum_{j=4}^{16} \binom{19-j}{2} \frac{\binom{20}{j}}{17} u_j.$$

The normalized derivatives are

$$G_j(X) = X^j - \binom{j}{3} X^{j-3} + \sum_{i=4}^j \binom{j}{i} u_i X^{j-i}, \quad 4 \leq j \leq 16.$$

For clarity, the integer presentation is

$$\begin{aligned} f_u = & X^{20} - 1140X^{17} + \sum_{j=4}^{16} C_j u_j X^{20-j} \\ & + \left(18221 - \sum_{j=4}^{16} (19-j) C_j u_j \right) X^2 + \left(-17082 + \sum_{j=4}^{16} (18-j) C_j u_j \right) X, \end{aligned} \quad C_j = \binom{20}{j}.$$

Direct substitution gives $f_u(1) = f'_u(1) = 0$, which defines $g_u = f_u/(X-1)^2$. The fixed reduction of g_u is

$$D = \frac{X^{20} - X^{17} - 3X^2 + 3X}{(X-1)^2} \quad \text{in } \mathbf{F}_{17}[X].$$

Its squarefreeness and factor degrees are checked in the exact presentation accompanying the paper.

Set $R_j = \text{Res}_X(g_u, G_j)$. The complete row-9 incidence problem is $R_4 = \dots = R_{16} = T = 0$ on integral parameters. The resultant equations without $T = 0$ are only a square enlargement.

The fixed reduction $D = \bar{g}_u$ is squarefree and has factor degrees 1, 1, 1, 5, 10. The simple roots are 0, 1, -2, five roots of Q_5 , and ten roots of Q_{10} , with Q_5, Q_{10} as in the input. Consequently all eighteen residue roots lie in $\mathbb{F}_{17^{10}}$. Let $K/\mathbb{Q}_{17}$ be the unramified extension of degree ten, $\mathcal{O}$ its integer ring, and $\pi = 17$.

The checked Frobenius identity is

$$r^{17} = \frac{3r}{r^2 + r + 1}.$$

For $r \neq 1$, putting $y = 7(r+1)/(r-1)$ gives

$$y^{17} = y^2 - 5.$$

These are residue-field identities. No exact characteristic-zero quadratic Frobenius equation is assumed.

6.2 A marked splitting lemma

Proposition 1. Let $\mathcal{O}$ be a complete discrete valuation ring with uniformizer π , and put $S = \mathcal{O}\langle u_1, \dots, u_m \rangle$, the restricted power-series ring. For each j choose $d_j \geq 1$ elements $L_{j,1}, \dots, L_{j,d_j}$ such that

$$\bar{L}_{j,r} = u_j + A_{j,r}(u_1, \dots, u_{j-1}). \quad (1)$$

Let $F_j = \prod_{r=1}^{d_j} L_{j,r}$ and $B = S/(F_1, \dots, F_m)$. Then:

1. B is finite free over $\mathcal{O}$, of rank $\prod_j d_j$.
2. For each marked tuple $\mathbf{r} = (r_1, \dots, r_m)$, $S/(L_{1,r_1}, \dots, L_{m,r_m})$ is a rank-one unital $\mathcal{O}$ -algebra and is therefore $\mathcal{O}$. Its coordinates give a point $u^{\mathbf{r}} \in \mathcal{O}^m$.
3. B has an S -module filtration with exactly these rank-one quotient modules, one for each marked tuple, including repetitions.
4. For every $H \in S$,

$$\det_{\mathcal{O}}(m_H | B) = \prod_{\mathbf{r}} H(u^{\mathbf{r}}). \quad (2)$$

Distinct marked tuples need not give distinct coordinate points. The proposition neither asserts that B is étale nor replaces B by its radical.

Proof. Consider any system obtained by replacing any of the F_j by a nonempty product of a subset of its factors. Its reduced equations, ordered by increasing j , are monic in successively fresh variables. They therefore form a regular sequence, and successive monic division gives a basis whose size is the product of the remaining factor counts.

The elementary lifting argument is as follows. If A is complete, separated and π -torsion-free and $\bar{h}$ is a nonzerodivisor in $A/\pi A$, then $A/(h)$ is π -torsion-free: an equation $\pi x = hy$ first gives $y = \pi z$, and cancellation gives $x = hz$. Also $hx = 0$ successively forces $x \in \pi^n A$ for every n , so h is a nonzerodivisor. Apply this argument to the reduced regular sequence. The resulting quotient is complete and torsion-free.

Lifting its finite monomial basis modulo π , successive approximation expresses every element as an $\mathcal{O}$ -linear combination of those same monomials. A relation among them has all coefficients divisible by π ; torsion-freeness permits division, and iteration makes the relation zero. Every full intermediate quotient is thus finite free with the stated rank. In particular each fully selected system has rank one. Its unit reduces to a basis of its one-dimensional residue algebra, so the structural map $\mathcal{O} \rightarrow S/(L_{1,r_1}, \dots, L_{m,r_m})$ is an isomorphism.

It remains to justify a filtration rather than just a rank count. Omit equation j , and call the resulting complete quotient A . Write the current product in that equation as ab , where a is one of its factors. The element $\bar{a}$ is a nonzerodivisor in $\bar{A}$. To see this, impose the lower-index equations first, obtaining a possibly nonreduced ring C . In $C[u_j]$, the element $\bar{a} = u_j + A_{j,r}$ is monic and is a nonzerodivisor. Adjoining each higher variable subject to its monic equation gives a finite free extension and preserves this property. The preceding lifting argument makes A torsion-free and a a nonzerodivisor. Consequently the following sequence is exact:

$$0 \longrightarrow A/(b) \xrightarrow{\cdot a} A/(ab) \longrightarrow A/(a) \longrightarrow 0. \quad (3)$$

Split each product recursively, then split the other equations. This produces the asserted filtration. Repeated factors are allowed in (3); they contribute repeated filtration factors.

Multiplication by H preserves the filtration. Each quotient is free of rank one over $\mathcal{O}$, and multiplication on it is the scalar $H(u^{\mathbf{r}})$. Choose an $\mathcal{O}$ -basis adapted to the filtration. The multiplication matrix is block triangular, so its determinant is the product in (2). $\square$

6.3 Consequences for the full row-9 algebra

In $\mathcal{O}\langle u_4, \dots, u_{16} \rangle[X]$, the polynomial g_u has a unique analytic root function $\rho_r(u)$ reducing to each residue root r of D . Indeed, a constant lift of r has g_u -value divisible by 17, while the derivative reduces to the nonzero constant $D'(r)$. The complete-ring Hensel iteration therefore constructs ρ_r . Distinct root functions have unit differences. Monicity and the degree give the exact factorization

$$g_u(X) = \prod_{r \in Z(D)} (X - \rho_r(u)).$$

Thus, without an omitted leading-coefficient factor,

$$R_j(u) = \prod_{r \in Z(D)} G_j(\rho_r(u)). \quad (4)$$

The reduction of a factor is

$$\overline{G_j(\rho_r(u))} = u_j + r^j - \binom{j}{3} r^{j-3} + \sum_{i=4}^{j-1} \binom{j}{i} u_i r^{j-i}.$$

It has precisely form (1). Proposition 1 applies with thirteen variables and eighteen factors per equation.

Corollary 2. After base change to $\mathcal{O}$, the completed row-9 resultant algebra has a filtration with 18^{13} rank-one factors. Its obstruction norm satisfies

$$N = \prod_{\mathbf{r} \in Z(D)^{13}} \tau_{\mathbf{r}}, \quad \tau_{\mathbf{r}} = T(u^{\mathbf{r}}) \in \mathcal{O}. \quad (5)$$

Every integral geometric solution of the square resultant system has all its coordinates in K . Every genuine row-9 CA polynomial, if one exists, has all its roots in K after this normalization.

For the last assertion, an integral solution in any valued extension chooses a vanishing factor of each product (4). The selected quotient is $\mathcal{O}$, so its parameters equal the corresponding $u^{\mathbf{r}}$. Then g_u splits into the displayed analytic roots in $\mathcal{O}$, and $f_u = (X-1)^2 g_u$ adds only the root one. This treats arbitrary ramification; it does not assume it away.

The same result applies to every fixed active support J , with zero removed from the domain and g_u/X in place of g_u . Its rank is $17^{|J|}$. Nominally active coefficients with zero residue remain included.

This fixed unramified field conclusion was already implicit in the marked Hensel proof of the existing batch exclusions. The additional point here is the filtration of the potentially nonreduced resultant algebra and the exact multiplicity-preserving norm identity (5).

Let σ be the Frobenius automorphism of $K/\mathbb{Q}_{17}$. Uniqueness of the marked lifts gives

$$u^{\sigma \mathbf{r}} = \sigma(u^{\mathbf{r}}), \quad \tau_{\sigma \mathbf{r}} = \sigma(\tau_{\mathbf{r}}). \quad (6)$$

The residue-domain cycles have lengths 1, 5, 10; hence so do marked tuple orbits. If an orbit has length d , its representative value belongs to the degree- d unramified subfield K_d , and its contribution to (5) is

$$\text{Norm}_{K_d/\mathbb{Q}_{17}}(\tau_{\mathbf{r}}).$$

If every factor is nonzero, then

$$v_{17}(N) = \sum_{\text{marked orbits } [\mathbf{r}]} d_{\mathbf{r}} v_{17}(\tau_{\mathbf{r}}). \quad (7)$$

All valuations in this sum are integers. This avoids any determinant cancellation issue, but not the need to establish nonvanishing on every relevant marked orbit.

6.4 Why the quadratic Frobenius law alone does not close the branch

The coefficients of the equations lie in $\mathbb{Z}_{17}$; the coefficients of an individual solution need not. Equation (6) transports a solution to its conjugate. It does not say that it fixes that solution. The following compact example shows the issue within the actual complete first-residue incidence equations.

Let r have irreducible polynomial

$$Q_5(r) = r^5 - 3r^4 - 2r^2 - 5r - 3 = 0 \quad \text{over } \mathbb{F}_{17}.$$

Use the active set $J = \{4, 9, 10, 14\}$ and witnesses

$$r_4 = -2, \quad r_9 = r, \quad r_{10} = 1, \quad r_{14} = -2.$$

At inactive indices choose zero. The derivative recurrences give

$$\begin{aligned} u_4 &= -7, \\ u_9 &= -5r^4 + 5r^3 + 8r^2 + 7r - 3, \\ u_{10} &= -r^4 + r^3 + 5r^2 - 2r + 4, \\ u_{14} &= -4r^4 + 4r^3 + 3r^2 - 8r - 4, \end{aligned}$$

and all other middle u_j vanish. Every marked witness is a root of $h = X^{20} - X^{17} - 3X^2 + 3X$, every marked normalized derivative vanishes, and direct reduction gives $\bar{T} = 0$. Nevertheless,

$$u_9^{17} - u_9 = 7r^4 + r^3 - 8r^2 - 7r + 7 \neq 0. \quad (8)$$

The last inequality follows from irreducibility and the nonzero degree-four remainder. The transformed $y = 7(r+1)/(r-1)$ also satisfies $y^{17} = y^2 - 5$, exactly as required.

Thus all first-residue constraints and the quadratic Frobenius law are consistent with non-Frobenius-fixed coefficients. Taking traces is legitimate, but cannot turn (6) into equality of the individual coefficient values. This is the already known degree-five residue fixture, here reconstructed symbolically; it is not a new surviving characteristic-zero candidate.

The residue obstruction itself has a compact expression:

$$\bar{T} = 4 + \sum_{j=4}^{16} \frac{3(-1)^j}{j(j-3)} u_j. \quad (9)$$

For $4 \leq j \leq 16$, cancelling the unique factor 17 in $\binom{20}{j}$ gives

$$\frac{\binom{20}{j}}{17} \equiv \frac{6(-1)^j}{j(j-1)(j-2)(j-3)} \pmod{17}.$$

Multiplication by $\binom{19-j}{2}$ proves (9). The weights are symmetric under $j \mapsto 20 - j$. Example (8) shows that this smaller formula is not itself a nonvanishing invariant.

6.5 Even the first three obstruction digits can vanish

There is a small exact example inside the true marked square system, not only a residue-field construction. Use $J = \{4, 5, 8, 10, 11\}$, initially put every active witness equal to one, and solve the normalized derivative equations recursively. This gives

$$(u_4, u_5, u_8, u_{10}, u_{11}) = (3, -6, 181, -7144, 50665).$$

All those normalized derivative incidences at one hold exactly, and

$$T = 11294240224 = 17^3 \cdot 2298848, \quad 17 \nmid 2298848. \quad (10)$$

For $q_u = g_u/X$, the exact identity $q_u(1) = 17T$ makes the single marked root equation vanish modulo 17^4 .

The derivative equations express the u_j as integer polynomials of the one common witness x . The marked root equation $q_{u(x)}(x) = 0$ has a unit derivative modulo 17: parameter dependence disappears modulo 17, and $\bar{q}'(1) \neq 0$. Its exact root therefore differs from one by an element of $17^4\mathbb{Z}_{17}$. All corresponding u_j , and hence T , change by multiples of 17^4 . Its exact square-system obstruction has valuation precisely three.

Consequently no uniform claim that all marked obstruction values are units, or have valuation at most one or two, can hold. This does not disprove a larger uniform bound. No such bound was established in this investigation.

7 Scope and remaining problem

The complete global statement proved here is a lower bound on the number of terms of a centered hypothetical counterexample in degree twenty. The accompanying support exclusions strengthen its finite frontier. They do not amount to a proof in degree twenty: denser supports and several reduction branches remain.

The local resultant factorization isolates a precise obstruction. Each marked square subsystem has a unique lift in a fixed unramified extension, while the extra scalar equation need not hold. To exclude the entire branch one must show that this scalar is nonzero for every relevant marking. The product formula does not accomplish that, and the examples show that several low-precision shortcuts fail. A uniform nonvanishing argument would be a substantive next result; enlarging a finite census is not a substitute for it.

The computational claims are reproducibility claims with explicit scope. Default replay verifies the recorded certificates and bounded checks; it does not rerun the largest optional native census. The original full-census receipts and separate arithmetic checks are retained. Compiler and timeout controls permit fresh runs on other systems, and a failed or interrupted run is not recorded as mathematical completion.

8 Data, code, and review provenance

The accompanying archive contains the editable manuscript, typesetting source, final PDF, proof notes, machine-readable certificates, source scripts, a file-integrity manifest, and a review response. Its README identifies the definitive proof and commands. The earlier consolidated report is retained as historical context; where its frontier counts differ, the explicitly scoped current support corollary governs only the newly excluded exact seven-term supports.

A supplied external review’s independent arithmetic script was rerun unchanged with the pure-Python SymPy ground backend; all eight reported sections matched. Its checks support the particular identities tested and do not replace proof of coverage, novelty assessment, or external refereeing of this revision. The new additions have separate internal review and replay records. File hashes establish integrity, not mathematical validity. This manuscript has not been submitted or published as part of preparing this review package.

9 Declarations for review

Data availability. The evidence and source code are included in the accompanying review archive; no permanent public repository or DOI is claimed.

Ethics. This is a mathematical investigation involving no human participants, animal experiments, or personal-data analysis.

Author contributions (CRediT), funding, and competing interests. Human authorship, CRediT assignments, funding information, and competing-interest declarations have not been supplied. These administrative declarations remain for the responsible author to complete before submission; absence of supplied information is not a declaration of no funding or no competing interests.

AI use. Codex assisted the mathematical investigation, code construction, literature comparison, drafting, and internal adversarial checks. Those internal checks are not human peer review or formal verification. The manuscript exposes the mathematical dependencies and executable evidence for independent assessment.

10 References

[CLO] Wouter Castryck, Robert Laterveer, Myriam Ounaies, *Constraints on counterexamples to the Casas-Alvero conjecture, and a verification in degree 12*. [arXiv:1208.5404v1](https://arxiv.org/abs/1208.5404v1), 27 August 2012; [full text](#). Relevant locations are Theorem 2 (degree $p + 1$, mean-root and determinant restrictions), Theorem 13 (distinct-root bound), Proposition 15 (prime-power derivative placement), and the scenario/descendant treatment in the computational sections. These are established predecessor methods, not campaign contributions. Their demonstrated full-degree computation is degree 12. In degree 20, derivative orders 1, 19, orders

4, 16, and the simultaneous triple 5, 10, 15 supply forbidden shared-root configurations. The appropriate hypotheses and normalization must be retained when translating these into coefficient conditions.

[deFrutos] Rosa María de Frutos Marín, *Perspectivas aritméticas para la Conjetura de Casas-Alvero*. Doctoral thesis, Universidad de Valladolid, repository year 2013; advisor Antonio Campillo López. [Primary PDF](#), [repository metadata](#), DOI [10.35376/10324/3602](#). The title-page date is 21 December 2012, the university catalog defense date is 11 June 2013, and repository availability is 27 September 2013; these describe distinct events. Sections 2.3 and 3.5, especially Proposition 3.5.5, contain the older sparse/two-visible-coefficient criterion used in the campaign’s support sieve. The full thesis text and Proposition 3.5.5 were retrieved and compared on 24 September 2026; see review/NOVELTY.md.

[Massri] Cesar Massri, *The Casas-Alvero conjecture for three recycled roots in degree 20*. [arXiv:1806.09561v6](#), [full text](#). First submission 25 June 2018; v6, 25 August 2023. Some earlier versions are marked withdrawn; the current v6 title and scope should be cited, not an earlier full-proof claim. Relevant passages are Remark 7.4, the proof of Theorem 7.9, Theorem 7.10, and the finiteness/normalization results in Sections 5–6. The restrictions on shared derivative pairs (5, 10), (10, 15) imply more sparsity than a title-only comparison reveals. Together with CLO they defeat novelty of the campaign’s initial five-term bound. The reported three-recycled-root computation is not a monomial-count theorem; no complete public replay of its reported 3^{17} assignments was retrieved. The current arXiv metadata supplies no journal reference.

[Marashdeh] Mohammad F. Marashdeh, *A descent-set obstruction for the Casas-Alvero conjecture*. [arXiv:2608.14726v1](#), [full text](#), 12 August 2026. The record exposes only v1. Relevant comparisons concern support stratification, triangular elimination of coefficients, descent-count determinants, and the two-element-support argument. These are not a degree-20 seven-total-term theorem. The campaign does not use a reported equation count as a proof that the remaining incidence system is empty. The preprint and its statements are distinct from external refereeing or a priority clearance.

[Bothmer-et-al] Hans-Christian Graf von Bothmer, Oliver Labs, Josef Schicho, Christiaan van de Woestijne, *The Casas-Alvero conjecture for infinitely many degrees*. [arXiv:math/0605090v2](#), 25 June 2007; first submission 3 May 2006. *Journal of Algebra* 316 (2007), 224–230, DOI [10.1016/j.jalgebra.2007.06.017](#). The weighted projective formulation and good-prime transfer, including Propositions 2.1–2.2 and 2.6, explain why geometric special-fibre emptiness can prove a characteristic-zero result and propagate degree n to np^e . Prime-field rational-point absence or an empty unsaturated affine chart does not meet those hypotheses.

[Berger] Laurent Berger, *The Weierstrass preparation theorem and resultants of p -adic power series*. [arXiv:1910.05319v2](#), 4 November 2019; first submission 11 October 2019. [Author-hosted PDF](#). Section 1, including Corollary 1.2, supplies a precise complete-ring preparation statement. In the campaign’s row-4 local argument, a power series reducing coefficientwise to z^2 over a complete unramified DVR is a unit times a monic distinguished quadratic. The campaign separately proves the hypotheses, convergence, and coverage of actual ramified points.

[ProofAtlas] [Casas–Alvero collaboration page](#). Reported work, inspected 24 September 2026; exact-system equivalence unresolved.

[Shih] Shih, Cheng-Pang, *On the Casas-Alvero Conjecture*, 2022 master’s thesis. [NTHU catalog record](#). Metadata inspected; full text unavailable.